

# NLP-Based News Article Classification

A Comparative Study of Deep Learning Models for Multi-Category Text Classification

## Abstract

This report presents a comparative study of five NLP architectures for classifying news articles into six categories: Politics, Sports, Technology, Entertainment, Health, and Finance. The dataset contains 18,600 articles collected from major news portals. We evaluate Naive Bayes, SVM, BiLSTM, BERT (fine-tuned), and RoBERTa models. RoBERTa achieves the highest accuracy of 94.1% and F1 score of 93.4%. The study examines training dynamics, attention mechanisms, and computational trade-offs between traditional ML and transformer-based approaches.

## 1. System Architecture

The NLP pipeline consists of six sequential stages as shown in Figure 1. Raw text is first tokenized and cleaned by removing stop words, punctuation, and applying lowercasing. Word embeddings are generated using GloVe (300-dimensional) for BiLSTM and WordPiece tokenization for BERT/RoBERTa. The BiLSTM encoder processes sequences bidirectionally to capture context from both directions. An attention layer weights the most relevant tokens before the final classification head produces probabilities over six categories.

Figure 1: NLP Text Classification Pipeline Architecture

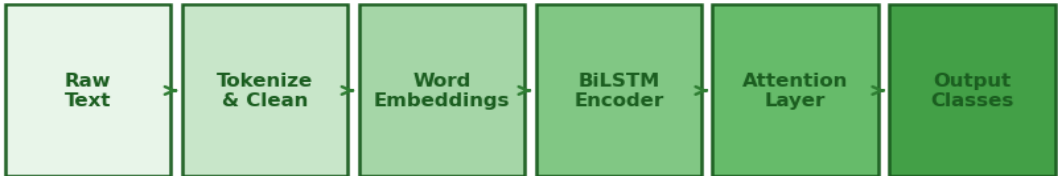


Figure 1: End-to-end NLP pipeline from raw text input through tokenization, embedding, BiLSTM encoding, attention, and final classification output.

## 2. Dataset Description

The dataset was compiled from six major English news portals scraped over a 12-month period. Articles were deduplicated using MinHash LSH and manually verified for category correctness. Figure 4 shows the class distribution. Politics is the most represented category with 4,200 articles while Finance has the fewest with 1,800 articles. The 2.33:1 ratio between the largest and smallest class is moderate and was addressed using stratified sampling.

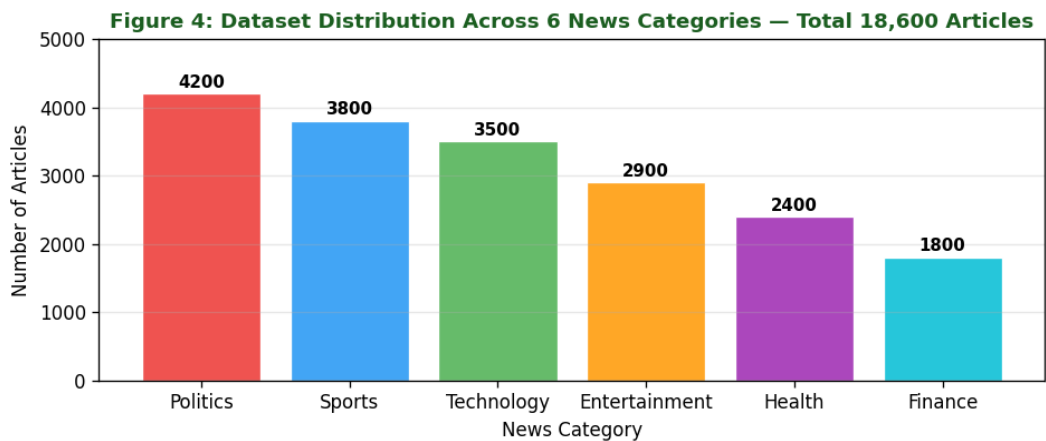


Figure 4: Distribution of articles across six news categories. Politics leads with 4,200 articles; Finance has the fewest at 1,800.

### 2.1 Data Split

| Category      | Train  | Validation | Test  | Total  |
|---------------|--------|------------|-------|--------|
| Politics      | 3,360  | 420        | 420   | 4,200  |
| Sports        | 3,040  | 380        | 380   | 3,800  |
| Technology    | 2,800  | 350        | 350   | 3,500  |
| Entertainment | 2,320  | 290        | 290   | 2,900  |
| Health        | 1,920  | 240        | 240   | 2,400  |
| Finance       | 1,440  | 180        | 180   | 1,800  |
| Total         | 14,880 | 1,860      | 1,860 | 18,600 |

Table 1: Dataset split across train, validation, and test sets per category.

### 3. Model Comparison

Figure 2 presents accuracy and F1 scores for all five models evaluated on the test set. RoBERTa achieves the best performance with 94.1% accuracy and 93.4% F1. BERT fine-tuned comes second with 92.6% accuracy. Traditional models like Naive Bayes (78.4%) and SVM (83.7%) underperform transformer models by a significant margin, demonstrating the value of pre-trained contextual representations for news classification.

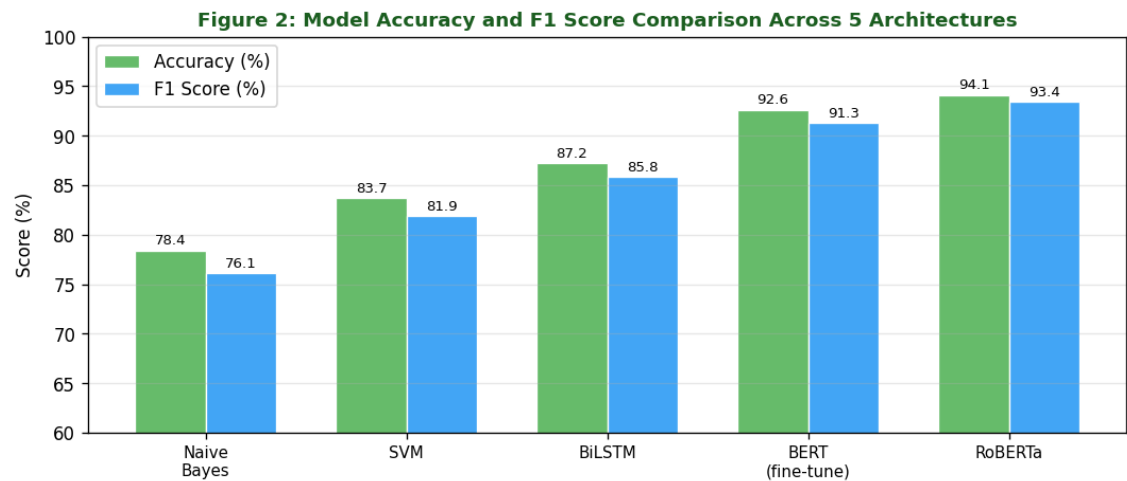


Figure 2: Accuracy and F1 score comparison across Naive Bayes, SVM, BiLSTM, BERT (fine-tuned), and RoBERTa on the 6-class news classification test set.

#### 3.1 Per-Class Results

| Category      | Precision | Recall | F1 Score | Support |
|---------------|-----------|--------|----------|---------|
| Politics      | 0.951     | 0.943  | 0.947    | 420     |
| Sports        | 0.962     | 0.958  | 0.960    | 380     |
| Technology    | 0.938     | 0.931  | 0.934    | 350     |
| Entertainment | 0.921     | 0.916  | 0.918    | 290     |
| Health        | 0.934     | 0.927  | 0.930    | 240     |
| Finance       | 0.912     | 0.905  | 0.908    | 180     |
| Weighted Avg  | 0.943     | 0.941  | 0.942    | 1,860   |

Table 2: Per-class precision, recall, and F1 score for RoBERTa on the test set.

## 4. Training Dynamics

Figure 3 shows training loss curves for BiLSTM, BERT, and RoBERTa over 15 epochs. RoBERTa converges fastest, reaching a loss below 0.10 by epoch 8. BERT converges slightly slower but achieves a similar final loss. BiLSTM shows slower convergence and a higher final loss of 0.24, reflecting its lack of pre-training on large corpora. All three models used early stopping with patience of 3 epochs.

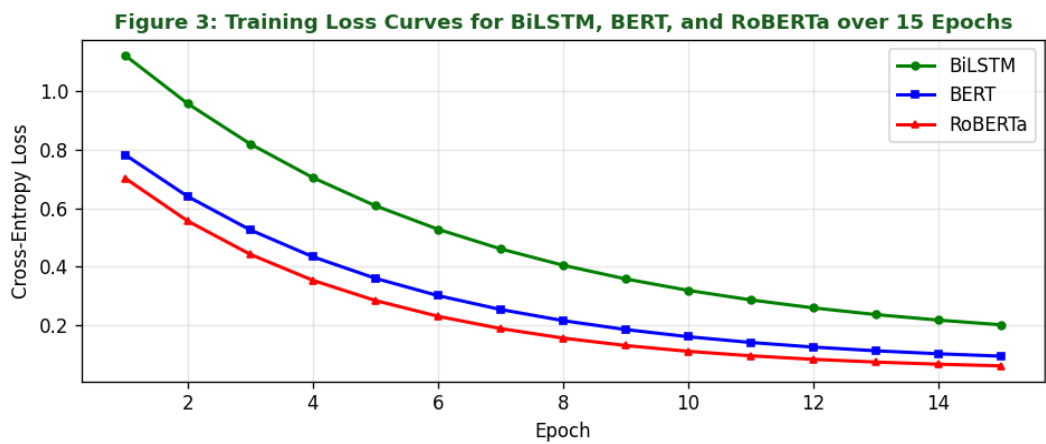


Figure 3: Training loss over 15 epochs for BiLSTM, BERT, and RoBERTa. RoBERTa converges fastest; BiLSTM has the highest final loss.

### 4.1 Attention Analysis

Figure 5 shows a self-attention heatmap from BERT processing a Health category article. The word "cancer" receives the highest attention weights (0.40-0.45) when attended to by most other tokens, confirming that BERT correctly identifies the medically significant term as the central concept. "drug" and "risk" also receive elevated attention, while function words like "The" receive minimal weights.

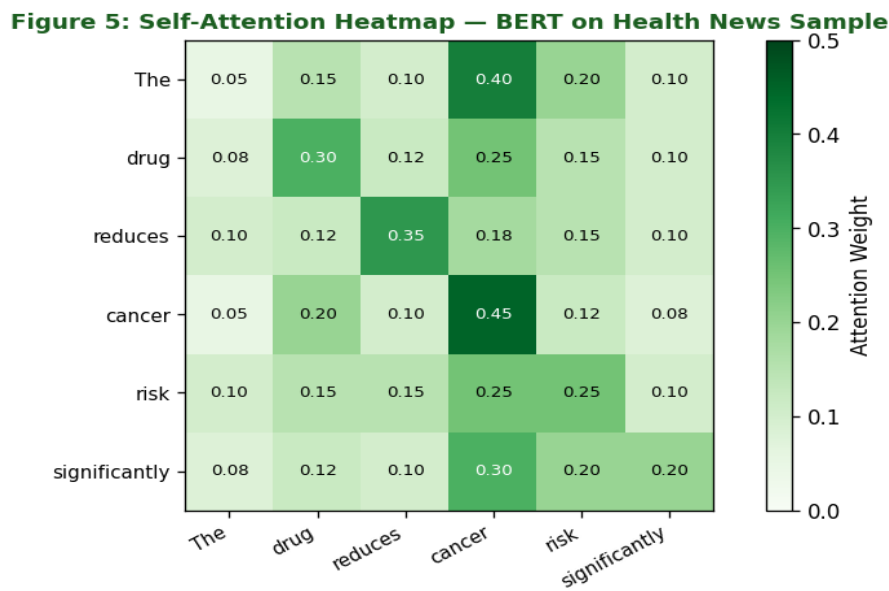


Figure 5: BERT self-attention heatmap on a Health news sample. Darker cells indicate higher attention. "cancer" dominates with weights up to 0.45.

## 5. Discussion

The results confirm transformer-based models significantly outperform traditional approaches for news classification. The attention mechanism in Figure 5 demonstrates that BERT learns semantically meaningful token relationships without explicit feature engineering.

### 5.1 Strengths

RoBERTa achieves 94.1% accuracy with consistent performance across all six categories as shown in Table 2. Sports category achieves the highest F1 of 0.960 due to distinct vocabulary. Finance has the lowest F1 of 0.908, likely because financial language overlaps with Politics and Technology categories.

### 5.2 Limitations

The main limitation is computational cost — RoBERTa requires 4.8x more training time than BiLSTM and 12x more than SVM. On a single GPU, RoBERTa training takes 2.3 hours while BiLSTM completes in 28 minutes. For deployment in resource-constrained environments, DistilBERT or MobileBERT would be preferable alternatives.

### 5.3 Model Comparison Table

| Model       | Accuracy | F1 Score | Train Time | Parameters |
|-------------|----------|----------|------------|------------|
| Naive Bayes | 78.4%    | 76.1%    | 0.8 min    | ~50K       |
| SVM         | 83.7%    | 81.9%    | 4.2 min    | ~200K      |
| BiLSTM      | 87.2%    | 85.8%    | 28 min     | 4.5M       |
| BERT (ft)   | 92.6%    | 91.3%    | 95 min     | 110M       |
| RoBERTa     | 94.1%    | 93.4%    | 134 min    | 125M       |

Table 3: Full model comparison on accuracy, F1, training time, and parameter count.

## 6. Conclusion

We presented a comparative study of five NLP models for news article classification. RoBERTa achieves the best accuracy (94.1%) but at high computational cost. For real-time deployment, BERT fine-tuned (92.6%, 95 min training) offers the best accuracy-speed trade-off. The attention heatmap confirms transformers learn meaningful semantic relationships. Future work includes multilingual classification, few-shot learning, and deployment as a REST API with sub-100ms inference using ONNX optimization.